Temporal Residual Map from the Completed Principal Symbol: Construction, Uniqueness, and the Lorentz Identification

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Abstract

The note [9] identifies the construction of a temporal residual map $\mathcal{P}_\tau^{\mathrm{Q5b}}$ as the remaining step needed to close the Lorentz identification from O7 observables. The present paper carries out this construction and proves that the map is the unique positive function compatible with three inputs simultaneously: the Lorentzian temporal splitting induced by the completed effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ of Q5b–Q11 [2, 13], the Born–Infeld unit capacity-flow sphere of [3], and the exact Weil linear filling law $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ of [9].

The completed principal symbol determines the Lorentzian temporal splitting of the effective tangent bundle; it does not by itself define a non-linear action on the scalar observable Σ_n^{tot}. The Born–Infeld admissibility budget provides the canonical unit capacity-flow lift, and the Weil filling law identifies the spatial component B_n. Their composition defines the scalar temporal residual map

$$\mathcal{P}_\tau^{\mathrm{Q5b}} \colon \Sigma_n^{\mathrm{tot}} \;\longmapsto\; \sqrt{\Sigma_n^{\mathrm{tot}}\big(2 - \Sigma_n^{\mathrm{tot}}\big)} \;=\; \sqrt{1 - B_n^2}.$$

This map is not an idempotent projection on scalar observables; it is the scalar shadow of a linear Lorentzian projection composed with the Born–Infeld lift. The three inputs are individually necessary: removing any one of them leaves the formula undetermined. This necessity retroactively explains the numerical failure of all three candidate maps tested in [9]. Lemma 1 of [9] follows, and with it Theorem 1: $\Phi(\eta_n^\tau) = (1 + (\eta_n^\tau)^2)^{-1/2} \to 1/\gamma$.

Keywords. temporal residual map, principal symbol, Born–Infeld admissibility, Lorentz factor, capacity-flow bundle, uniqueness, Heisenberg group, sub-Riemannian geometry, effective co-metric, Weil factorization, Cosmochrony.

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1 Introduction

1.1 Context and the open problem

The Cosmochrony spectral geometry programme derives effective spacetime structure from admissibility constraints on non-injective projections of spectral data on the Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ [1]. The paper Q5b [2] constructs the effective operator $L_{\mathrm{eff}} = -A_\tau \partial_\tau^2 + A_H \Delta_H$ on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$, extracts a Lorentzian co-metric from its principal symbol, and leaves A_τ open. Papers Q8 [10], Q10 [11], and Q11 [13] close this gap by proving $A_z = A_H = A_\tau = 2$, yielding the completed effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ with no free parameter.

The note [9] then studies the O7 saturation observable $\eta_n^{O7} = B_n/\Sigma_n^{\mathrm{tot}}$ and asks whether it recovers the relativistic Lorentz factor γ. Two structural facts are established there:

1. The Weil factorization gives $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ exactly in the Schrödinger representation (Corollary 1 of [9]).

2. A temporal residual projection $\mathcal{P}_\tau^{\mathrm{Q5b}}$ is needed; no natural BFS observable directly satisfies the quadratic law $B_n^2 + (\Sigma_n^\tau)^2 = 1$.

The note defines $\mathcal{P}_\tau^{\mathrm{Q5b}}$ implicitly via the completed co-metric of Q5b and Q11, but defers its explicit construction as "the remaining open task" (Section 6.3 of [9]).

1.2 What the present paper does

We carry out this construction in full and prove that the resulting map is the *unique* positive function compatible with the Lorentzian splitting, the BI budget, and the Weil filling law (Theorem 4.5). The uniqueness is the core result: once the three inputs are specified, the formula is forced.

As a by-product, the uniqueness proof explains why all three candidate maps Σ_n^A, Σ_n^B, Σ_n^C tested numerically in [9] fail: each violates exactly one of the three necessary inputs (Proposition 5.2).

1.3 Precise scope of each ingredient

The derivation rests on three distinct inputs. Each is necessary; removing any one leaves the formula undetermined.

(L) **Lorentzian splitting** (from the completed principal symbol): the co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ determines a unique timelike direction ∂_τ in $T(\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R}))$ and a corresponding linear projector P_τ on vectors. The principal symbol alone does not determine a scalar map on Σ_n^{tot}.

(BI) **Born–Infeld capacity sphere** (from the BI admissibility budget [3]): the admissible capacity flow vector $\vec{F}_n$ lies on the unit sphere $(F_n^\tau)^2 + |\vec{F}_n^{\,\mathrm{spat}}|^2 = 1$ (after normalisation). This is the quadratic constraint that forces $F_n^\tau = \sqrt{1 - B_n^2}$ once B_n is known.

(W) **Weil filling law** (from the Schrödinger factorization [9]): $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ exactly, identifying the spatial component of $\vec{F}_n$ as $B_n = 1 - \Sigma_n^{\mathrm{tot}}$.

The temporal residual map is the unique positive function $\psi\colon [0,1] \to [0,1]$ compatible with all three.

1.4 Organisation

Section 2 fixes notation and recalls the relevant results. Section 3 defines the capacity-flow bundle and the unit BI sphere. Section 4 constructs $\mathcal{P}_\tau^{\mathrm{Q5b}}$ and proves Theorem 4.5. Section 5 proves the necessity of each ingredient and explains the failure of the three candidates of [9]. Section 6 derives the Lorentz identification. Section 7 gives the logical status table.

2 Setup and Notation

2.1 Objects in force

Throughout, q denotes a prime, $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and d_w denotes the BFS word metric on G_q. The O7 cumulative occupation observable at depth $n < q/2$ (proved exact in [9]) is

$$B_n \;:=\; \frac{2n^2+2n+1}{q^2}, \qquad n < q/2. \tag{1}$$

This is the fraction of (a,b) pairs in $(\mathbb{Z}/q\mathbb{Z})^2$ satisfying $\min(a, q-a) + \min(b, q-b) \le n$, i.e. the normalised spatial footprint of the BFS ball at depth n. It satisfies $B_0 = 1/q^2$, $B_n < 1/2$ for $n < q/2$, and is strictly monotone in the pre-wrapping window $n < q/2$. The formula (1) does not extend beyond $n = q/2$; the full saturation profile on the finite torus involves wrapping corrections. The O7 total residual frontier novelty is the exact complement (Corollary 1 of [9]):

$$\Sigma_n^{\mathrm{tot}} \;=\; 1 - B_n. \tag{2}$$

2.2 Completed co-metric and its inverse

The effective operator on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ is (Q5b Theorem 5.2 [2]):

$$L_{\mathrm{eff}} = -A_\tau \partial_\tau^2 + A_H \Delta_H, \tag{3}$$

where $\Delta_H = \tilde{X}^2 + \tilde{Y}^2$ is the Kohn–Laplacian on $\mathrm{Heis}_3(\mathbb{R})$. The principal symbol of L_{eff} at (p,k) is:

$$\sigma_2(L_{\mathrm{eff}})(p,k) = -A_\tau k_\tau^2 + A_H\Big[\Big(k_x + \tfrac{y}{2}k_z\Big)^2 + \Big(k_y - \tfrac{x}{2}k_z\Big)^2\Big]. \tag{4}$$

Following Q5b [2] (equations (17)–(19)), the effective *co-metric tensor* $g^{\mu\nu}$ is read off from the principal symbol by $\sigma_2(L_{\mathrm{eff}})(p,k) = g^{\mu\nu}(p)k_\mu k_\nu$. We denote by $g_{\mu\nu}$ its inverse metric tensor (used for lowering indices and computing Lorentzian projections); these are related by $g^{\mu\rho}g_{\rho\nu} = \delta^\mu_\nu$.

The following results are in force without further proof: Q8 [10] gives $A_z = 2$; Q10 [11] gives $A_H = 2$ under [U]; Q11 [13] gives $A_\tau = 2$ under the Q5a hypotheses and [U]. The completed effective co-metric is therefore

$$g^{\mu\nu} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1,+1,+1,+1), \tag{5}$$

with no free parameter, and its inverse metric is

$$g_{\mu\nu} = \tfrac{1}{2}\eta_{\mu\nu}. \tag{6}$$

The temporal component satisfies $g_{\tau\tau} = -1/2 < 0$. Formulae for Lorentzian projections in Section 4 use $g_{\mu\nu}$ (6).

2.3 Born–Infeld admissibility budget

The Born–Infeld admissibility constraint of [3] bounds the total admissible capacity increment per projective depth step dn:

$$(dI_\tau)^2 + (dI_{\mathrm{spat}})^2 = c_{\mathrm{BI}}^2\, dn^2. \tag{7}$$

After normalisation $c_{\mathrm{BI}} = 1$, the capacity flow vector $\vec{F}_n$ (Definition 3.1 below) lies on the unit Born–Infeld sphere. This budget is a structural axiom of the Foundation [1]; it is not derived from the principal symbol.

3 Capacity-Flow Bundle and Born–Infeld Sphere

3.1 The capacity flow vector

In the large-q continuum limit, the cascade depth n is identified with the coordinate τ on $\mathbb{R}_\tau$ (Q5b Remark 3.3 [2], Q11 Lemma 3.1 [13]). The O7 observables become sections of a bundle over $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$.

Definition 3.1 (Capacity flow vector and unit BI sphere)**.** *The* capacity flow vector *at depth n is*

$$\vec{F}_n := F_n^\tau\, \partial_\tau + \vec{F}_n^{\,\mathrm{spat}} \;\in\; T_{(\tau,e)}(\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})),$$

where $F_n^\tau = dI_\tau/dn$ and $|\vec{F}_n^{\,\mathrm{spat}}| = dI_{\mathrm{spat}}/dn$ are the normalised temporal and spatial capacity increments. The BI budget (7) *places $\vec{F}_n$ on the* unit Born–Infeld sphere*:*

$$(F_n^\tau)^2 + |\vec{F}_n^{\,\mathrm{spat}}|^2 = 1. \tag{8}$$

3.2 Identification of the spatial component

The O7 observable B_n is the normalised spatial coverage of the BFS cascade. By the Q5b Carnot–Carathéodory convergence (Theorem 3.2 of [2]), in the large-q limit B_n coincides with the fraction of spatial capacity occupied by the BFS ball, i.e. the magnitude of the spatial component of the normalised capacity flow:

$$B_n = |\vec{F}_n^{\,\mathrm{spat}}|. \tag{9}$$

Substituting (9) into the unit BI sphere (8) gives

$$F_n^\tau = \sqrt{1 - B_n^2}. \tag{10}$$

4 Construction and Uniqueness of $\mathcal{P}_\tau^{\mathrm{Q5b}}$

4.1 The linear Lorentzian projector on vectors

The completed co-metric (5) endows $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ with a Lorentzian structure. The associated *temporal projector* on tangent vectors is the standard Lorentzian orthogonal projection onto the timelike direction ∂_τ:

$$P_\tau V := \frac{g(V, \partial_\tau)}{g(\partial_\tau, \partial_\tau)}\, \partial_\tau = \frac{g_{\tau\mu} V^\mu}{g_{\tau\tau}}\, \partial_\tau = V^\tau\, \partial_\tau, \tag{11}$$

using $g_{\tau\tau} = -1/2$ from (6) and $g_{\tau\mu} V^\mu = -V^\tau/2$. The map $V \mapsto P_\tau V = V^\tau \partial_\tau$ is *linear and idempotent* on $T(\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R}))$.

Remark 4.1 (Scope of the linear projector)**.** *The principal symbol determines P_τ via the co-metric* (5)*. It does not, however, determine a map on the scalar observable $\Sigma_n^{\mathrm{tot}} \in [0,1]$: P_τ acts on tangent vectors, not on real-valued capacity sections. The lift from scalars to vectors is provided by the BI bundle structure (Definition 3.1), not by the principal symbol.*

4.2 The scalar temporal residual map

Definition 4.2 (Temporal residual map)**.** *The* temporal residual map $\mathcal{P}_\tau^{\mathrm{Q5b}}$ *is the composition:*

$$\mathcal{P}_\tau^{\mathrm{Q5b}}\colon \Sigma_n^{\mathrm{tot}} \xrightarrow{\textit{BI lift}} \vec{F}_n \xrightarrow{P_\tau} F_n^\tau\,\partial_\tau \xrightarrow{\textit{magnitude}} |F_n^\tau|.$$

Explicitly, $\mathcal{P}_\tau^{\mathrm{Q5b}}(\Sigma_n^{\mathrm{tot}}) := |F_n^\tau|$*, where* F_n^τ *is the temporal component of the capacity flow vector (Definition 3.1).*

Remark 4.3 (Non-idempotence)**.** *The map* $\mathcal{P}_\tau^{\mathrm{Q5b}}\colon [0,1]\to[0,1]$ *is* not *idempotent:* $(\mathcal{P}_\tau^{\mathrm{Q5b}})^2 \neq \mathcal{P}_\tau^{\mathrm{Q5b}}$ *in general. It is not an algebraic projection on a space of scalar observables. The name "temporal residual map" is preferred over "projector" for the scalar action; the underlying linear map* P_τ *on tangent vectors is an orthogonal projector.*

Remark 4.4 (Central direction)**.** *As noted in [9] (Section 5.3),* $\mathcal{P}_\tau^{\mathrm{Q5b}}$ *does not act on the Schrödinger representation space* $\mathbb{C}^q$*. The central direction* $\tilde{Z} = \partial_z$ *of* $\mathrm{Heis}_3(\mathbb{R})$ *becomes the temporal direction only after the symbol completion chain CC structure → sub-principal correction (Q8) → Q11 closure →* $g^{\mu\nu} = 2\eta^{\mu\nu}$*. The map* $\mathcal{P}_\tau^{\mathrm{Q5b}}$ *is defined at the level of the completed geometry on* $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$*, not at the level of the Weil representation.*

4.3 Main theorem: explicit formula and uniqueness

Theorem 4.5 (BI-lifted temporal residual map)**.** *Assume the Q5a continuum hypotheses [6]* [H1], [H-w], [H-E1], [C]*, the spectral universality hypothesis* [U] *proved in U1 [14], and the canonical BI capacity-flow lift of Definition 3.1. The temporal residual map* $\mathcal{P}_\tau^{\mathrm{Q5b}}$ *acts on the O7 total residual* $\Sigma_n^{\mathrm{tot}} \in [0,1]$ *by*

$$\mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}} \;=\; \sqrt{\Sigma_n^{\mathrm{tot}}\big(2-\Sigma_n^{\mathrm{tot}}\big)} \;=\; \sqrt{1-B_n^2}. \tag{12}$$

Consequently, the quadratic law

$$B_n^2 \;+\; \big(\mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}}\big)^2 \;=\; 1 \tag{13}$$

holds exactly for all $n < q/2$*.*

Moreover, $\mathcal{P}_\tau^{\mathrm{Q5b}}$ *is the* unique *map* $\psi\colon [0,1]\to[0,1]$ *satisfying simultaneously:*

(L) $\psi(\Sigma_n^{\mathrm{tot}}) = |P_\tau \vec{F}_n|$ *(Lorentzian projection of the capacity flow);*

(BI) $B_n^2 + \psi(\Sigma_n^{\mathrm{tot}})^2 = 1$ *(unit BI sphere* (8)*);*

(W) $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ *(Weil filling law* (2)*).*

Proof. **Explicit formula.** By the Weil filling law (2): $B_n = 1 - \Sigma_n^{\mathrm{tot}}$. By the spatial identification (9) and the unit BI sphere (8):

$$(F_n^\tau)^2 = 1 - B_n^2 = 1 - (1-\Sigma_n^{\mathrm{tot}})^2 = \Sigma_n^{\mathrm{tot}}(2-\Sigma_n^{\mathrm{tot}}).$$

By Definition 4.2: $\mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}} = |F_n^\tau| = \sqrt{\Sigma_n^{\mathrm{tot}}(2-\Sigma_n^{\mathrm{tot}})}$. Equation (13) follows immediately: $B_n^2 + (1-B_n^2) = 1$. This identity is exact for all $n < q/2$.

Uniqueness. Let $\psi\colon [0,1]\to[0,1]$ satisfy (L), (BI), and (W) simultaneously. By (W): $B_n = 1 - f$ where $f = \Sigma_n^{\mathrm{tot}}$. Substituting into (BI): $(1-f)^2 + \psi(f)^2 = 1$, hence $\psi(f)^2 = 1-(1-f)^2 = f(2-f)$. The unique positive branch is $\psi(f) = \sqrt{f(2-f)}$. No other positive function satisfies all three conditions. □

Remark 4.6 (Derivation chain)**.** *The three conditions (L), (BI), (W) are jointly necessary and sufficient. The Lorentzian splitting (L) specifies* which *component of* $\vec{F}_n$ *is temporal (the* ∂_τ *direction, with negative g-norm). The BI sphere (BI) fixes the normalization of* $\vec{F}_n$ *to unity, forcing* $(F_n^\tau)^2 = 1 - B_n^2$ *once* B_n *is known. The Weil filling law (W) provides the identification* $B_n = 1 - \Sigma_n^{\mathrm{tot}}$*, converting the formula to a function of the observable alone. Dropping any one of these three leaves the formula undetermined.*

5 Necessity of Each Ingredient

5.1 What fails without each ingredient

Proposition 5.1 (Necessity)**.** *Each of conditions* (L)*,* (BI)*,* (W) *is individually necessary for the uniqueness in Theorem 4.5.*

- *Without* (L)*: the temporal component of $\vec{F}_n$ is unspecified; any component of $\vec{F}_n$ could play the role of Σ_n^τ.*
- *Without* (BI)*: the capacity flow vector is not constrained to the unit sphere; $(F_n^\tau)^2 + B_n^2$ can take any value in $(0, \infty)$, and the quadratic law* (13) *need not hold.*
- *Without* (W)*: the spatial component B_n is not identified with $1 - \Sigma_n^{\text{tot}}$; the formula for F_n^τ as a function of Σ_n^{tot} alone is then inaccessible.*

Proof. Direct: each condition is used exactly once in the proof of Theorem 4.5. Without (L), the map $V \mapsto P_\tau V$ is undefined, so Definition 4.2 has no content. Without (BI), equation (8) does not hold, and $|F_n^\tau|$ cannot be expressed in terms of B_n alone. Without (W), the substitution $B_n = 1 - \Sigma_n^{\text{tot}}$ is unavailable, and ψ cannot be expressed as a function of Σ_n^{tot} alone. □

5.2 Retroactive explanation of the candidate failures

The note [9] (Section 6) tests three candidate maps Σ_n^A, Σ_n^B, Σ_n^C and finds that all fail the quadratic law $B_n^2 + (\Sigma_n^X)^2 \to 1$ with mean deviations of 45%, 13%, and 7% respectively. Proposition 5.1 now gives the structural reason.

Proposition 5.2 (Failure modes)**.** *The three candidate maps of [9] fail the quadratic law because each violates exactly one of conditions* (L)*,* (BI)*,* (W)*.*

1. *$\Sigma_n^A = 1 - |c : \exists(a,b),\, (a,b,c) \in \bigcup_{k \le n} S_k|/q$ (central-coordinate coverage): fails* (W)*. The central variable c is not the complement of B_n; it saturates at depth $n \approx q/2$ while $B_n \approx 1/2$, so $\Sigma_n^A \neq 1 - B_n$. This observable measures central BFS coverage, a distinct quantity from the total Weil residual.*
2. *$\Sigma_n^B = \sqrt{1 - B_n}$ (square-root complement): fails* (BI)*. It satisfies $B_n + (\Sigma_n^B)^2 = 1$ but not $B_n^2 + (\Sigma_n^B)^2 = 1$. This is the identity arising from a linear budget $B_n + \Sigma = 1$ with Σ quadratically normalized, not from the BI sphere.*
3. *$\Sigma_n^C = \sqrt{1 - \beta_n^2}$ with $\beta_n = n/n_{\max}$ (depth proxy): fails* (W)*. The normalised depth $\beta_n = n/n_{\max}$ is not equal to $B_n = (2n^2 + 2n + 1)/q^2$ (they differ by a quadratic-vs-linear discrepancy in n), so the Weil identification $B_n = 1 - \Sigma_n^{\text{tot}}$ is not satisfied for Σ_n^C.*

Proof. For Σ_n^A: the Weil factorization proves $\Sigma_n^{\text{tot}} = 1 - B_n$ exactly ([9] Corollary 1), but Σ_n^A measures the distinct observable of central-coordinate BFS coverage, which is not equal to $1 - B_n$ in the mid-range of the cascade. For Σ_n^B: $B_n + (\sqrt{1 - B_n})^2 = B_n + 1 - B_n = 1$, confirming it satisfies the *linear* complement law, not the BI sphere $B_n^2 + (\Sigma_n^B)^2 = 1$. For Σ_n^C: $\beta_n = n/n_{\max}$ while $B_n = (2n^2 + 2n + 1)/q^2$; these satisfy $B_n \approx \beta_n^2 \cdot n_{\max}^2/q^2$, a different quadratic relationship, so (W) fails for Σ_n^C. □

Remark 5.3 (Informational content)**.** *The failure of Σ_n^C by only 7% is now explained: $\beta_n \approx B_n^{1/2}$ at intermediate depths (since $B_n \propto n^2/q^2$), so $\Sigma_n^C = \sqrt{1 - \beta_n^2} \approx \sqrt{1 - B_n}$, which is close to $\sqrt{1 - B_n^2}$ when B_n is small but diverges for $B_n \to 1$, consistent with the reported residuals near $\beta_n \approx 0.8$.*

6 The Lorentz Identification

Theorem 4.5 proves Lemma 1 of [9]. Theorem 1 of [9] then applies.

Corollary 6.1 (Lorentz factor)**.** *Under the hypotheses of Theorem 4.5, the temporally projected saturation ratio*

$$\eta_n^\tau := \frac{B_n}{\mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}}} = \frac{B_n}{\sqrt{1-B_n^2}} = \beta\gamma,$$

where $\beta := B_n$ is the normalised spatial mobility and $\gamma = (1-\beta^2)^{-1/2}$ is the Lorentz factor. Consequently,

$$\Phi(\eta_n^\tau) := \bigl(1+(\eta_n^\tau)^2\bigr)^{-1/2} = \frac{1}{\gamma}.$$

Proof. Direct from $\mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}} = \sqrt{1-B_n^2}$ and the definition of γ. □

Remark 6.2 (Physical content)**.** *Corollary 6.1 gives: (i)* time dilation*: $d\tau/dn = 1/\gamma$; (ii)* light cone*: $B_n = 1$ forces $F_n^\tau = 0$, i.e. no temporal residual at $v = c$. Length contraction and the reconstruction of effective Lorentz transformations between inertial observers are addressed in [8].*

7 Logical Status

7.1 Status table

The table below lists the status of every ingredient.

Label	Content	Status
[9] Cor. 1	$\Sigma_n^{\mathrm{tot}} = 1 - B_n$ exact	Proved (Weil factorization)
Q5b Thm. 5.2 [2]	L_{eff} and principal symbol	Proved (Q9 [12] + Q5a)
Q8 [10]	$A_z = 2$	Proved (Casimir rigidity)
Q10 [11]	$A_H = 2$	Proved (conditional on [U])
Q11 [13]	$A_\tau = 2$, $g^{\mu\nu} = 2\eta^{\mu\nu}$	Proved (conditional on [U])
BI budget (7) [3]	Unit BI sphere	Structural (Foundation [1])
Definition 3.1	Capacity flow vector $\vec{F}_n$	Structural (this paper)
Theorem 4.5	Formula and uniqueness	Proved (conditional on [U] + BI lift)
Corollary 6.1	$\Phi(\eta_n^\tau) = 1/\gamma$	Proved (conditional on [U] + BI lift)

The sole analytic conditional is hypothesis [U], proved in U1 [14] under the Q5a hypotheses. The BI lift (Definition 3.1) is structural: it uses the BI budget axiom [3, 1] and the Q5b spatial identification.

7.2 Closures of LC-O1–O3

LC-O1 (Length contraction). Effective Lorentz transformations between inertial observers are reconstructed in [8]. The Born–Infeld capacity sphere supplies the load β and the residual $F^\tau = 1/\gamma$, while the Q5b–Q11 metric closure supplies the Lorentzian interval whose isometries are the Lorentz group. Length contraction follows as the reciprocal spatial projection onto the simultaneity hyperplane of the moving splitting.

LC-O2-O1 (Capacity–metric bridge). The equivalence $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$, postulated in Remark 3.2 of [4], is established in [7]. The identity follows from local admissibility-weight reduction on $\mathrm{Sym}^2(V_\rho)$, without using the Einstein equation.

LC-O3 (Finite-q corrections at saturation). Finite-q corrections near saturation are analysed in [5]. The quadratic identity $B_n^2 + (F_n^\tau)^2 = 1$ is exact for every n with the exact torus load. The corrections affect the parabolic Carnot identification $B_n^{\mathrm{parab}} = (2n^2 + 2n + 1)/q^2$, not the capacity identity itself: for $n = R + m$,

$$B_n^{\mathrm{torus}} = B_n^{\mathrm{parab}} - \frac{4m^2}{q^2}.$$

The parabolic identification is exact for $n \le R := (q-1)/2$ and asymptotically exact for $n = R + o(q)$.

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